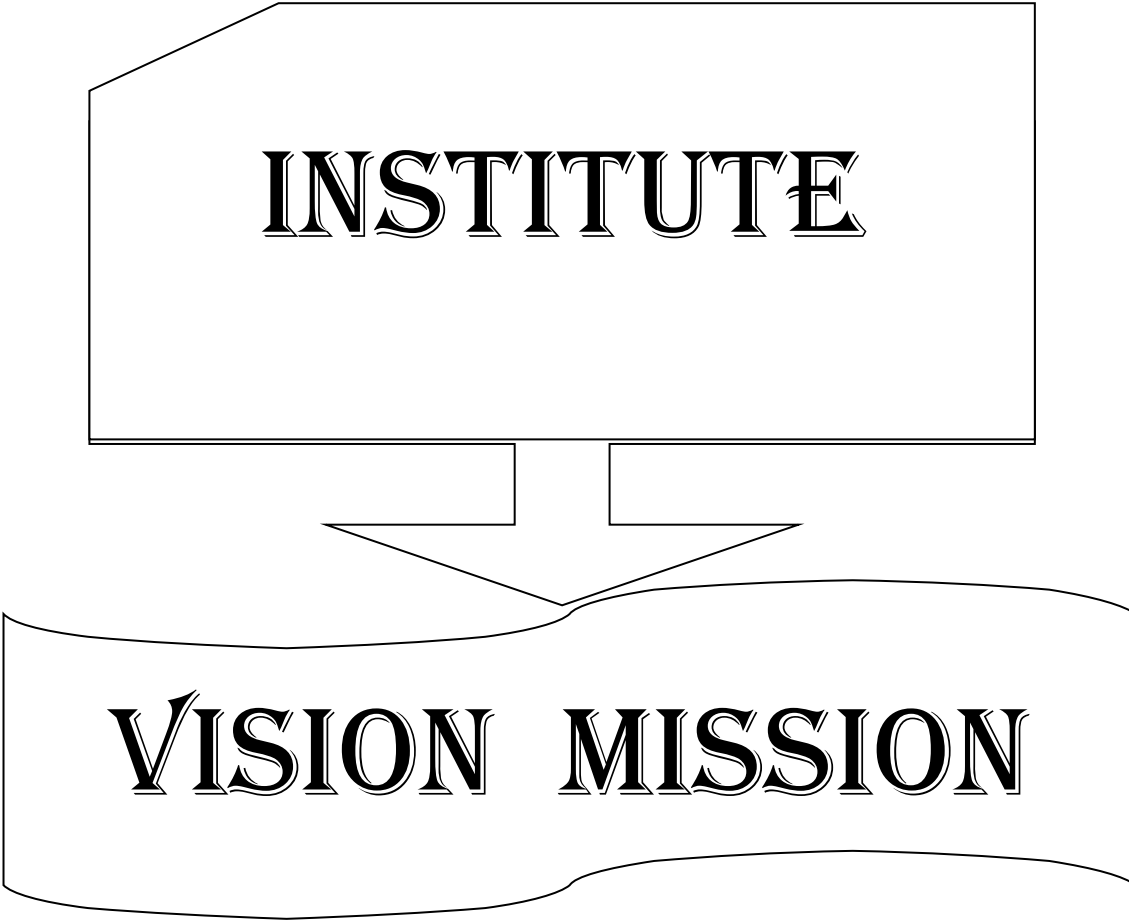


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INSTITUTE

VISION MISSION

INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.

DEPARTMENT

VISION

MISSION

DEPARTMENT VISION AND MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs)

Graduates of this programme will :

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSO s):

A graduate of the Computer Science and Engineering Program will be able to:

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. **[K3]**
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. **[K4]**
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. **[K5]**
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. **[K5]**
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. **[K3]**
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. **[K3]**
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. **[K3]**
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. **[K3]**
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. **[K6]**
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. **[K2]**
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. **[K6]**
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. **[K1]**

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.as@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, **TADEPALLIGUDEM** – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 25-03-2021

Academic Calendar for B.Tech IV & VI Semesters **Academic Year 2020-21**

B.Tech IV & VI Semesters			
Description	From	To	Weeks
Commencement of Class Work	12.04.2021		
I Unit of Instructions	12.04.2021	29.05.2021	7 W
I Mid Examinations	24.05.2021	29.05.2021	1 W
II Unit of Instructions	31.05.2021	17.07.2021	7 W
II Mid Examinations	12.07.2021	17.07.2021	1 W
Comprehensive Examinations	19.07.2021	24.07.2021	1 W
Preparation & Practicals	26.07.2021	31.07.2021	1 W
End Examinations	02.08.2021	14.08.2021	2 W
Commencement of Next Semester Class Work	16.08.2021		


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: B. Tech. VI SEM

Sections: A & B

w.e.f :03.05.2021

Periods	1	2	3	4
Time	(10:00AM-11:00 AM)	(11:30 PM-12:30 PM)	(02.00 PM-03.00 PM)	(03:30 PM-04.30 PM)
Date & Day				
Monday	PCS-IV	ML/STM	CD	TT
Tuesday	CD	IOT	DM	TT
Wednesday	C&NS	DM	OOAD	TT
Thursday	PCS-IV	OOAD	ML/STM	TT
Friday	ML/STM	DM	C&NS	TT
Saturday	IOT	C&NS	CD	TT

Class: B. Tech. VI SEM

Sections: C & D

Periods	1	2	3	4
Time	(10:00AM-11:00 AM)	(11:30 PM-12:30 PM)	(02.00 PM-03.00 PM)	(03:30 PM-04.30 PM)
Date & Day				
Monday	CD	ML/STM	PCS-IV	TT
Tuesday	PCS-IV	IOT	DM	TT
Wednesday	C&NS	DM	OOAD	TT
Thursday	DM	OOAD	ML/STM	TT
Friday	ML/STM	C&NS	CD	TT
Saturday	IOT	CD	C&NS	TT

Staff Details:

S. No.	Course Code	Course Name	Section	
			A & B	C & D
1.	V18CST19	Compiler Design (CD)	Mrs. P. Suneetha	Mrs. D. Anjani Suputri Devi
2.	V18CST20	Data Mining (DM)	Dr. D. Jaya Kumari / Mr. CH. Hemanand	Dr. G. Loshma
3.	V18CST21	Object Oriented Analysis and Design through UML (OOAD)	Ms. M. Madhavi	Mrs. S Krishnapriya
4.	V18CST22	Cryptography & Network Security (CN&S)	Mrs. N. Hiranmayee	Dr. V.S.Naresh
5.	V18CST23	Professional Elective-II Software Testing Methodologies (STM)	Mr. N.V.Murali Krishna Raja	Mr. N.V.Murali Krishna Raja
6.	V18CST25	Professional Elective-II Machine Learning (ML)	Mr. David Raju. K	Mr. S. Kumar Reddy Mallidi
7.	V18ECTO E1	Open Elective – I Internet of Things (IOT)	Mr. T D N S Sarveswara Rao	Mr. K S Surya Kiran
8.	V18ENI06	Professional Communication Skills – IV (Aptitude) PCS-IV	Mrs. K V L B Devi / Mr. T Suresh	
		Professional Communication Skills – IV (Verbal) PCS-IV	Mr. J N V Somayajulu / Mr. P. Someswara Rao	
9.	V18CST63	Technical Skills-IV	Mr. L Balaji	
10.		Technical Training	Dr. V Venkateswara Rao	

Head of the Department



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

Class: VI Semester
Section: A

Class Coordinator: Ms. M. Madhavi

w.e.f. 15.04.2021
Room No : G-201

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	ML/STM	DM	PCS-IV (Verbal)	IOT	Lunch Break	Technical Training / TS-IV		
Tue	OOAD LAB		DM LAB			PCS-IV (Aptitude)		CNS
Wed	DM	Technical Training				ML/STM	CD	CNS
Thu	CD	IOT	DM	CNS		OOAD	MPS LAB	
Fri	OOAD	CNS	DM	ML/STM		Technical Training		DM(T)
Sat	CD	ML/STM	IOT	OOAD		IOT	OOAD	CD

Section : B

Class Coordinator :Mrs. P. Suneetha

Room No : G-202

Periods	1	2	3	4	1:00P M 2:00P M	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DM	IOT	ML/STM	PCS-IV (Verbal)	Lunch Break	Technical Training / TS-IV		
Tue	CD	CNS	ML/STM	DM		PCS-IV (Aptitude)		OOAD
Wed	ML/STM	Technical Training				OOAD	DM	IOT
Thu	OOAD LAB		DM LAB			ML/STM	IOT	DM
Fri	CNS	CD	CD	OOAD		Technical Training		DM(T)
Sat	OOAD	CNS	CD	IOT		CNS	MPS LAB	

Section :C

Class Coordinator :Mrs.SKrishnapriya

Room No : G-203

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	ML/STM	PCS-IV (Aptitude)		IOT	Lunch Break	PCS-IV (Verbal)	DM	OOAD
Tue	DM	CD	CNS	OOAD		Technical Training		
Wed	OOAD LAB		DM LAB			ML/STM	CNS	IOT
Thu	OOAD	Technical Training / TS-IV				DM	CD	CNS
Fri	IOT	CD	CNS	ML/STM		IOT	MPS LAB	
Sat	CD	ML/STM	OOAD	DM		Technical Training		
Sun	MPS LAB							

Section :D

Class Coordinator :Mr. N.V.Murali Krishna Raja

Room No: G-204

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	DM	CNS	ML/STM	OOAD	Lunch Break	PCS-IV (Aptitude)		IOT
Tue	CNS	OOAD	ML/STM	DM		Technical Training		
Wed	ML/STM	PCS-IV (Verbal)	IOT	CNS		CD	MPS LAB	
Thu	CD	Technical Training / TS-IV				ML/STM	IOT	OOAD
Fri	OOAD LAB		DM LAB			DM	CD	CNS
Sat	OOAD	DM	IOT	CD		Technical Training		
Sun	Technical Training							

Staff Details:

S. No.	Course Code	Course Name	Section			
			A	B	C	D
11.	V18CST19	Compiler Design (CD)	Mrs. P. Suneetha		Mrs. D. AnjaniSuputri Devi	
12.	V18CST20	Data Mining (DM)	Dr. D. Jaya Kumari	Mr. CH. Hemanand	Dr. G. Loshma	
13.	V18CST21	Object Oriented Analysis and Design through UML (OOAD)	Ms. M. Madhavi		Mrs.SKrishnapriya	
14.	V18CST22	Cryptography & Network Security (CN&S)	Mrs. N. Hiranmayee		Dr. V.S.Naresh	
15.	V18CST23	Professional Elective-II Software Testing Methodologies (STM)	Mr. N.V.Murali Krishna Raja		Mr. N.V.Murali Krishna Raja	
16.	V18CST25	Professional Elective-II Machine Learning (ML)	Mr. David Raju. K		Mr. S. Kumar Reddy Mallidi	
17.	V18ECTO E1	Open Elective – I Internet of Things (IOT)	Mr.T D N S SSarveswara Rao		Mr.K S Surya Kiran	
18.	V18CSL08	Object Oriented Analysis and Design through UML Lab (OOAD LAB)	Ms. M. Madhavi / Mr. David Raju. K		Mrs.SKrishnapriya / Mrs. D. AnjaniSuputri Devi	Mrs.SKris hnapriya / Mr. David Raju.
19.	V18CSL09	Data Mining Lab (DM LAB)	Dr. D. Jaya Kumari/ Mrs. N. Hiranmayee/ Mrs. P. Suneetha	Mr. CH. Hemanand / Mr. S. Kumar Reddy Mallidi	Dr. G. Loshma/ Mr. S. Kumar Reddy Mallidi	Dr. G. Loshma/ Mrs. N. Hiranmayee
20.	V18CSMPS	Mini Project with Seminar(MPS)	Ms. M. Madhavi	Mrs. N. Hiranmayee	Mr. S. Kumar Reddy Mallidi	Mrs.SKri shnapriya
21.	V18ENI06	Professional Communication Skills –IV(Aptitude) PCS-IV	Mrs.K V L B Devi / Mr.T Suresh			
		Professional Communication Skills –IV(Verbal) PCS-IV	Mr.J N V Somayajulu/Mr.P.Someswara Rao			
22.	V18CST63	Technical Skills-IV	Mr. L Balaji			
23.		Technical Training	Dr.V Venkateswara Rao			

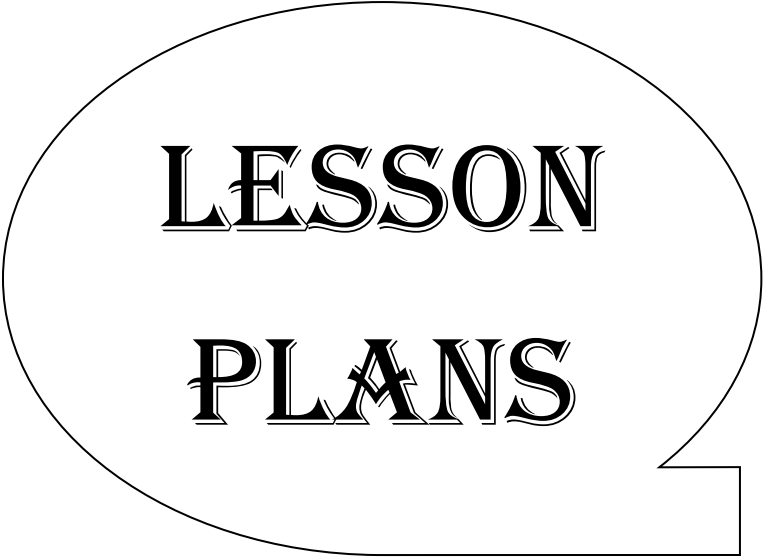

 Head of the Department

COURSE STRUCTURE

VI Semester

VI - Semester							
S.No.	Course Code		Course	L	T	P	C
1	V18CST19	PCC	Compiler Design	3	0	0	3
2	V18CST20	PCC	Data Mining	3	0	0	3
3	V18CST21	PCC	Object Oriented Analysis and Design through UML	3	0	0	3
4	V18CST22	PCC	Cryptography & Network Security	3	0	0	3
5	Elective - II			3	0	0	3
	V18CST23	PEC	1. Software Testing Methodologies				
	V18CST24		2. Principles of Programming Languages				
	V18CST25		3. Machine Learning				
	V18CST26		4. Image Processing				
6	Open Elective – I (Interdisciplinary)	OEC	OPE I(1-3)	3	0	0	3
7	V18CSL08	PCC	Object Oriented Analysis and Design through UML Lab	0	0	3	1.5
8	V18CSL09	PCC	Data Mining Lab	0	0	3	1.5
9	V18CSMPS	Mini Project	Mini Project	0	0	4	2
10	V18ENT05		Professional Communication Skills - III	4	0	0	MN C
11	V18CST63		Technical Skills-IV	4	0	0	MN C
Total				26	0	10	23

Total Contact Hours: 36



LESSON PLANS

Compiler Design

LESSON PLAN

Academic Year: 2020-21

Year/ Semester: VI

Name of the Course: Compiler Design

Programme: B.Tech

Section: A,B,C & D

Course Code: V18CST19

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	COURSE OUTCOMES (After completion of the course, The Learner is able to)	KNOWLEDGE LEVEL
CO1	Describe the compilation process and lexical analyzer	K2
CO2	Construct top down parsing Techniques	K3
CO3	Construct bottom up parsing techniques	K3
CO4	Construct syntax directed translation	K3
CO5	Produce intermediate code generation process and run time environments	K3
CO6	Explain the code generation process.	K2

Text Books:

1. Compilers, Principles Techniques and Tools- Alfred V Aho, Monica S Lam, Ravi Sethi, Jeffrey D. Ullman, 2nd ed, Pearson, 2007.

REFERENCES

1. Principles of compiler design, V. Raghavan, 2nd ed, TMH, 2011

2. Compiler Design, K. Muneeswaran, Oxford

Targeted Proficiency Level and Attainment level (For each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	55	55	55	55	55	55
	Level 2	50	50	50	50	50	50
	Level 1	45	45	45	45	45	45

UNIT – I

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
		Dissemination of Vision, Mission, PEOs,POs,PSOs		1		PPT
1.	CO1	Describe Language Processors	K1	1	Lecture with discussion	PPT
2.		Explain the Structure of a Compiler	K2	2	Lecture with discussion	PPT
3.		Lexical Analysis: Explain The Role of the Lexical Analyzer	K2	2	Lecture with demonstration	PPT
4.		Define Specification of Tokens	K1	1	Lecture with demonstration	PPT
5.		Describe Recognition of Tokens	K1	1	Lecture with demonstration	PPT
6.		Construct Lexical-Analyzer Generator Lex	K2	2	Lecture with demonstration	PPT
Total Hours				10 Hours		

UNIT – II

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO2	Define CFG	K1	1	Lecture with discussion	PPT
		Explain Lexical Versus Syntactic Analysis	K2	1	Lecture with discussion	PPT
2.		Explain Elimination of Left Recursion	K2	1	Lecture with discussion	ICT
3.		Construct Left Factoring	K2	1	Lecture with discussion	ICT
4.		Construct Top Down Parsing: Recursive Descent Parsing	K3	2	Lecture with discussion	ICT
5.		Construct First and Follow	K3	2	Lecture with demonstration	ICT
		ConstructLL(1) Grammars	K3	3	Lecture with demonstration	ICT
6.		Construct Non recursive Predictive Parsing	K3	2	Lecture with demonstration	PPT
7.		Demonstrate Error Recovery in Predictive Parsing.	K3	1	Lecture with demonstration	PPT
Total Hours				14Hours		

UNIT – III

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO3	Explain Bottom Up Parser Classification	K2	1	Lecture with discussion	PPT
2.		Explain Reductions, Handle Pruning	K2	1	Lecture with discussion	PPT
		Construct Shift-Reducing	K3	2	Lecture with demonstration	ICT
5.		Explain Conflicts During Shift Reduce Parsing	K2	1	Lecture with discussion	ICT
6.		Define LR Parsing	K1	1	Lecture with demonstration	ICT
7.		Explain Difference between LR and LL Parsers	K2	1	Lecture with demonstration	PPT
9.		Construct LR Parsers,Items and the LR(0) automaton	K2	3	Lecture with demonstration	ICT
10.		ExplainThe LR-Parsing Algorithm	K2	1	Lecture with demonstration	ICT
11.		Construct SLR Parsing Tables	K2	2	Lecture with demonstration	ICT
Total Hours				13 Hours		

UNIT – IV

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO4	Construct CLR (1)	K3	2	Lecture with demonstration	ICT
2.		Construct LALR Parsing tables	K3	2	Lecture with demonstration	ICT
3.		Complete Comparison of all Bottom Up approaches	K3	1	Lecture with demonstration	PPT
		Define Semantic Analysis: Syntax Directed Definitions	K1	1	Lecture with demonstration	PPT
		Explain Evaluation Orders for SDD's	K2	2	Lecture with demonstration	PPT
		Explain Applications of SDT	K2	1	Lecture with demonstration	PPT
Total Hours				09 Hours		

UNIT – V

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO5	Explain Variants of Syntax Trees	K2	1	Lecture with discussion	PPT
		Explain Three-Address Code	K2	1	Lecture with discussion	PPT
2.		Demonstrate Control Flow	K3	1	Lecture with discussion	PPT
		Explain Back-patching	K2	1	Lecture with discussion	PPT
3.		Explain Run-Time Environments: Storage Organization	K2	1	Lecture with demonstration	PPT
		Illustrate Stack Allocation of Space	K3	1	Lecture with discussion	PPT
		IllustrateHeap Management	K3	1	Lecture with discussion	PPT
Total Hours				07 Hours		

UNIT – VI

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO6	Explain Basic Blocks and Flow Graphs	K2	1	Lecture with discussion	PPT
2.		Describe Optimization of Basic Blocks	K2	1	Lecture with discussion	PPT
		Explain Peephole Optimization	K2	1	Lecture with discussion	PPT
3.		Explain Register Allocation and Assignment	K2	1	Lecture with discussion	PPT
4.		Illustrate Machine-Independent optimizations: The Principal Sources of Optimizations	K2	2	Lecture with demonstration	PPT
5.		Discuss Introduction to Data-Flow Analysis.	K2	1	Lecture with demonstration	PPT
Total Hours				07 Hours		

Total No. of classes: 60

Data Mining

LESSON PLAN

Academic Year: 2020-21

Year/ Semester: VI

Name of the Course: Data Mining

Programme: B.Tech

Section: A,B,C& D

Course Code: V18CST20/313

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome
1	313.1	Explain the concept of Data Mining and its functionalities.[K2]
2	313.2	Discuss various Data Preprocessing Techniques [K2]
3	313.3	Demonstrate Association Analysis Techniques. [K3]
4	313.4	Illustrate various Classification Techniques.[K3]
5	313.5	Demonstrate Alternative techniques for Classification [K3]
6	313.6	Use different Clustering techniques to cluster data.[K3]

Text Books:

1. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers

Reference Books:

1. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

2. Data Mining and Analysis, Mohammed J Zaki, Wagner Meira JR, 1st Edition, Cambridge University Press

Lecture Plan

UNIT - I

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1.	CO1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs, COs, Introduction: Identify what motivated Data Mining.	K1	1	Lecture	BB/ICT
2.		State the importance of Data Mining. Describe kinds of Data on which Data Mining can be done.	K1	2	Lecture	BB/ICT
3.		Illustrate Data Mining Functionalities.	K2	2	Lecture+ discussion	BB/ICT
4.		Illustrate Major Issues in Data Mining	K1	1	Lecture	BB/ICT
5.		Explain Attribute Types and Basic Statistical Descriptions of Data	K2	2	Lecture + discussion	BB/ICT
6.		Illustrate Data Visualization techniques.	K2	1	Lecture + discussion	BB/ICT
7.		Describe Data Similarity and Dissimilarity measures	K2	2	Lecture + discussion	BB/ICT
		Total		11		

UNIT - II

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	CO2	Data Pre-processing: Identify reasons for pre-processing the data.	K1	1	Lecture	BB/ICT
2.		Describe Descriptive Data Summarization.	K2	1	Lecture + discussion	BB/ICT
3.		Explain Data Cleaning techniques.	K2	2	Lecture + discussion	BB/ICT
4.		Discuss Data Integration and Data Transformation techniques	K2	2	Lecture + discussion	BB/ICT
5.		Explain Data Reduction techniques.	K2	2	Lecture + discussion	BB/ICT
6.		Discuss Data Discretization and Concept Hierarchy Generation.	K2	2	Lecture + discussion	BB/ICT
		Total		10		

UNIT - III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Reqd.	Pedagogy	Teaching aids
1.	C03	Illustrate the concept of Association Analysis	K2	2	Lecture + discussion	BB/ICT
2.		Explain Frequent Itemset generation process.	K2	2	Lecture + discussion	BB/ICT
3.		Discover Association Rules from the given data.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
4.		Present a Compact representation of frequent itemsets.	K1	1	Lecture	BB/ICT
5.		Use FP-Growth Algorithm for Association Analysis.	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
		Total		9		

UNIT - IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req'd.	Pedagogy	Teaching aids
1.	CO4	Classification : Describe the basic concepts of Classification	K1	1	Lecture	BB/ICT
2.		Illustrate the approach for solving a classification problem.	K2	1	Lecture + discussion	BB/ICT
3.		Explain the concept of a Decision Tree Induction	K2	2	Lecture + discussion	BB/ICT
4.		Construct a decision tree.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
5.		Describe the methods for expressing an attribute test conditions.	K2	1	Lecture + discussion	BB/ICT
6.		Identify the measures for selecting the best split.	K2	1	Lecture + discussion	BB/ICT
7.		Illustrate Tree Pruning	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
		Total		09		

UNIT - V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req'd.	Pedagogy	Teaching aids
1.	CO5	Classification : Alternative techniques Explain Bayes' Theorem	K2	2	Lecture+ discussion	BB/ICT
2.		Apply Naïve Bayesian Classification Algorithm	K3	2	Lecture + discussion and In-class Assignment	BB/ICT
3.		Explain the concept of Bayesian Belief Networks	K2	2	Lecture + discussion	BB/ICT
		Total		06		

UNIT – VI

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Req'd.	Pedagogy	Teaching aids
1.	CO6	Explain the Concept of Cluster Analysis.	K2	1	Lecture + discussion	BB/ICT
2.		Describe different types of Clustering and Different Types of Clusters.	K2	2	Lecture + discussion	BB/ICT
3.		Apply k-means and its variants for clustering.	K3	3	Lecture + discussion and In-class Assignment	BB/ICT
4.		Describe k-means additional issues.	K1	1	Lecture	BB/ICT
5.		Describe k-means and different types of clusters.	K2	1	Lecture + discussion	BB/ICT
6.		Identify k-means strengths and weaknesses	K1	1	Lecture	BB/ICT
7.		Express k-means as an Optimization Problem.	K2	1	Lecture + discussion	BB/ICT
8.		Apply Basic Agglomerative Hierarchical Clustering.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
9.		Discuss other Specific Techniques for Agglomerative Hierarchical Clustering	K2	1	Lecture + discussion	BB/ICT
10.		Explain Traditional Density: Center-Based Approach.	K2	1	Lecture + discussion	BB/ICT
11.		Apply the DBSCAN Algorithm.	K3	1	Lecture + discussion and In-class Assignment	BB/ICT
12.		List strengths and Weaknesses of DBSCAN Algorithm	K1	1	Lecture	BB/ICT
		Total		15		

OOAD THROUGH UML

LESSON PLAN

Academic Year: 2020-21

Year/ Semester: VI

Name of the Course: OOAD THROUGH UML Course Code: V18CST21/CS314

Programme: B.Tech

Section: A,B,C& D

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BT L
1.	V18CST21.1	Discuss importance of modeling.	K2
2.	V18CST21.2	Describe classes and relationships.	K2
3.	V18CST21.3	Develop class diagrams and object diagrams.	K3
4.	V18CST21.4	Develop Interaction, Use case and Activity Diagrams.	K3
5.	V18CST21.5	Illustrate advanced behavioral modeling.	K3
6.	V18CST21.6	Develop component and deployment diagrams	K3

TEXT BOOK:

1. The Unified Modeling Language User Guide, Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education.

REFERENCE BOOKS:

1. UML 2 Toolkit, Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, WILEY- Dreamtech India Pvt. Ltd.
2. Fundamentals of Object Oriented Design in UML, Meilir Page-Jones, Pearson Education.
3. Modeling Software Systems Using UML2, Pascal Roques, WILEY- Dreamtech India Pvt. Ltd. Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		75	60	75	60	60	75
Targeted level of Attainment	Level 3	75	60	75	60	60	75
	Level 2	65	50	65	50	50	65
	Level 1	55	40	55	40	40	55

UNIT-1

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs, POs, PSOs		1	Lecture	BB
2		Introduction to UML: Introduction	K1	1	Lecture	BB
3		Importance of modeling - Principles of modeling	K1	1	Lecture	ICT
4		Object oriented modeling, Introduction to UML	K2	1	Lecture	ICT
5		Conceptual model of the UML	K2	2	Lecture	ICT+ BB
6		Architecture	K2	1	Lecture	ICT
7		Software Development Cycle	K2	1	Lecture	ICT
8		Total		8		

UNIT-2

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Advanced Structural Modeling : Classes	K2	1	Lecture	BB
2		Relationships	K2	1	Lecture	BB
3		Common Mechanisms and Diagrams	K2	2	Lecture	ICT
4		Advanced classes	K2	1	Lecture	ICT+ BB
5		Advanced relationships	K2	1	Lecture	ICT+ BB
6		Interfaces	K2	1	Lecture	ICT+ BB
7		Types and Roles – Packages.	K2	1	Lecture	BB
9		Total		8		

UNIT-3

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Class & Object Diagrams: Terms, concepts	K2	2	Lecture	BB+ICT
2		Modeling techniques for Class Diagrams	K3	2	Lecture	BB+ICT
3		Modeling techniques for Object Diagrams.	K3	2	Lecture	BB+ICT
4		Total		6		

UNIT-4

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Basic Behavioral Modeling-I: Interactions	K2	2	Lecture	BB+ICT
2		Interaction diagrams.	K3	2	Lecture with Discussion	BB+ICT
3		Basic Behavioral Modeling-II: Use cases	K2	2	Lecture	BB+ICT
4		Use case Diagrams	K3	2	Lecture with Discussion	BB+ICT
5		Activity Diagrams	K3	3	Lecture with Discussion	BB+ICT
6		Total		11		

UNIT-5

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Advanced Behavioral Modeling: Events and signals	K ₂	1	Lecture	BB+ICT
2		State machines	K ₂	1	Lecture	BB+ICT
3		Processes and Threads	K ₂	1	Lecture	BB+ICT
4		Time and space	K ₂	2	Lecture	BB+ICT
5		State chart diagrams	K ₃	3	Lecture	BB+ICT
6		Total		9		

UNIT-6

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Architectural Modeling: Component	K ₂	2	Lecture	BB+ICT
2		Deployment	K ₂	2	Lecture	BB+ICT
3		Component diagrams	K ₃	2	Lecture	BB+ICT
4		Deployment diagrams.	K ₃	2	Lecture	BB+ICT
5		Total		8		

Total No. of Classes : 60

Cryptography & Network Security

Academic Year: 2019-20

Year/ Semester: III/II

Name of the Course: Cryptography & Network Security

Programme: B.Tech

Section: A,B , C & D

Course Code: V18CST22

COURSE OUTCOMES (Along with Knowledge Level):

After completion of this course, the students will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	V18CST22.1	Describe the fundamentals of networks security, security architecture, threats and vulnerabilities.	K2
2.	V18CST22.2	Discuss the mathematical support for both symmetric and asymmetric key cryptography.	K2
3.	V18CST22.3	Discuss the concept of developing encryption and decryption algorithms.	K2
4.	V18CST22.4	Illustrate various techniques of encryption and message authentication functions.	K3
5.	V18CST22.5	Apply various Key management and Distribution techniques and its importance.	K3
6.	V18CST22.6	Demonstrate the Need of Transport level and Electronic mail security algorithms.	K3

TEXT BOOKS:

1. William Stallings, “Cryptography and Network Security, Principles and Practices”, Pearson Education, Sixth Edition.
2. William Stallings, “Network Security Essentials (Applications and Standards)”, Pearson Education Fourth Edition.
3. Cryptography and Network Security, Behrouz A Forouzan, Debdeep Mukhopadhyay, (3e) McGrawHill.

REFERENCE BOOKS:

1. Charlie Kaufman, Radia Perlman and Mike Speciner, “Network Security – Private Communication in a Public World” Pearson/PHI.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:**UNIT-1**

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission, PEOs,POs,PSOs		1	Lecture	ICT
2		Explain Computer security concepts	K2	1	Lecture	ICT
3		Classify Security Attacks	K2	1	Lecture	ICT
4		Explain about Security Services and Mechanisms	K2	1	Lecture with Discussion	ICT
5		Explain OSI Security Architecture	K2	2	Lecture	ICT
6		Explain Model for Network Security	K2	1	Lecture	ICT
7		Classify Classical Encryption Techniques	K2	1	Lecture with Discussion	ICT
8		Classify Substitution ciphers	K2	2	Lecture	ICT
9		Classify Transposition Techniques	K2	2	Lecture	ICT
10		Total		12		

UNIT-2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Describe Prime and Relatively Prime Numbers	K1	1	Lecture	ICT
2		Illustrate about Fermat's and Euler's Theorems	K3	2	Lecture	ICT
3		Explain The Chinese Remainder theorem	K2	2	Lecture	ICT
4		Explain Euclidean Algorithm	K2	1	Discussion	ICT
5		Explain Modular Arithmetic	K2	2	Discussion	ICT
6		Total		8		

UNIT-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Define Block Cipher	K1	1	Lecture	ICT
2		Describe Traditional Block Cipher Structure	K2	1	Lecture	ICT
3		Explain about Stream Cipher	K2	1	Lecture	ICT
4		Explain about RC4 Stream Cipher	K2	1	Lecture	ICT
5		Explain about DES	K2	2	Discussion	ICT
6		Explain Triple DES	K2	1	Discussion	ICT
7		Explain about AES-Structure	K2	2	Lecture	ICT
8		Describe Key Expansion	K1	1	Lecture	ICT
9		Discuss IDEA	K2	1	Lecture	ICT
10		Total		11		

UNIT-4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Illustrate RSA Algorithms	K3	1	Lecture	ICT
2		Illustrate Diffie Hellman Key Exchange	K3	1	Lecture	ICT
3		Discuss Elliptic Curve Cryptography	K2	1	Lecture	ICT
4		Explain about Message Authentication code - Message Authentication Functions	K2	1	Lecture	ICT
5		Explain about Requirements and security for Message Authentication code	K2	2	Lecture With Discussion	ICT
6		Explain HMAC and Hash functions	K2	2	Lecture With Discussion	ICT
7		Explain Secure Hash	K2	1	Lecture	ICT

		Algorithm(SHA)				
8		Explain SHA-512	K2	2	Discussion	ICT
9		Total		11		

UNIT-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Explain about DigitalSignatures	K2	1	Lecture	ICT
2		Explain Authentication Protocols	K2	2	Lecture	ICT
3		Explain about Kerberos	K3	2	Lecture	ICT
4		Differentiate about Key management distribution(Symmetric &Asymmetric)	K3	1	Lecture	ICT
5		Explain X.509 digital certificate	K3	2	Lecture	ICT
6		Explain about NIST DigitalSignature Algorithm	K2	1	Lecture	ICT
7		Total		9		

UNIT-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Transport Level Security Discuss Web security Requirements	K2	1	Lecture With Discussion	ICT
2		Explain about Secure Socket Layer (SSL)	K2	2	Lecture	ICT
3		Explain Transport Layer Security(TLS)	K2	1	Discussion	ICT
4		Illustrate principleofInclusion–Exclusion	K3	1	Lecture	ICT
5		Explain Electronic Mail Security: Pretty Good Privacy (PGP)	K2	2	Lecture	ICT

6		Demonstrate about S/MIME	K3	2	Lecture	ICT
9		Total		9		

Total No. of Classes: 60

Software Testing Methodologies

Academic Year: 2020-21

Programme: B.Tech

Year/ Semester: VI

Section: A,B,C,D

Name of the Course: Software Testing Methodologies **Course Code:** V18CST23

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	1	Describe Software testing objectives and methodology.	K2
2.	2	Apply various Software testing techniques.	K3
3.	3	Discuss Static testing techniques for software testing.	K2
4.	4	Differentiate software testing and debugging process.	K2
5.	5	Construct test cases by understanding test suite management.	K3
6.	6	Explain modern software testing tools to support software testing.	K2

Text Books:

1. Software testing Principles and Practices, Naresh Chouhan, 9th Edition, Oxford Publisher

Reference Books:

1. Software testing techniques - Boris Beizer, 2nd Edition, Dreamtech publisher..
2. Foundations of Software testing, Aditya P Mathur, 2nd edition, Pearson.
3. Software Testing- Yogesh Singh, cambridge.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		65	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

Unit-1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept.and PEOs,POs & PSOs of the Programme.		1	Lecture	BB
2		Describe Evolution of software testing.	K1	1	Lecture with discussion	BB+ICT
3		Identify Myths and Facts, Goals of software testing.	K1	1	Lecture	BB
4		Outline the Model for software testing.	K1	1	Lecture	BB
5		Describe the life cycle of a bug and its states ,Affect of bugs on software testing economics	K1	1	Lecture	BB+ICT
6		Classify bugs based on criticality and SDLC	K2	1	Lecture	BB
7		Explain Software Testing life cycle.	K2	2	Lecture with Discussion and in class Assignment	BB+ICT

Total 8

Unit- 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Outline Verification and Validation activities	K1	1	Lecture	BB
2		Explain the Verification of requirements.	K2	1	Lecture	BB
3		Interpret the process to verify High level and Low level designs.	K2	1	Lecture	BB
4		Explain about code verification and Validation activities.	K2	1	Lecture	BB
5		Define Black box testing and list techniques.	K1	1	Lecture	BB
6		Apply Boundary Value Analysis for creating test cases.	K3	2	Lecture with Discussion	BB+ICT
7		Demonstrate Equivalence Class testing technique	K3	1	Lecture with Discussion	BB+ICT

8		Illustrate Decision table based testing technique.	K3	2	Lecture with Discussion and in class Assignment	BB+ICT
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Total 10

Unit-3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define White Box Testing and criteria for Logic Coverage.	K1	1	lecture	BB
2		Illustrate Basis Path testing .	K2	2	lecture	BB+ICT
3		Explain various Loops and their Testing strategies.	K2	2	lecture	BB+ICT
4		Define Static Testing and its objectives	K1	1	lecture	BB
5		Explain about software Inspections.	K2	2	lecture	BB
6		Describe the technique of Structured Walkthroughs.	K2	1	Lecture with Discussion and in class Assignment	BB
7		Explain about Technical Reviews.	K2	1	lecture	BB

Total 10

Unit- 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Define Regression testing and State when it is done.	K1	1	Lecture	BB
2		Describe the necessity of regression testing and its objectives.	K2	1	Lecture with Discussion and in class Assignment	BB
3		Explain Regression testing Techniques	K2	2	Lecture	BB+ICT
4		Describe the steps in Debugging Process	K1	1	Lecture	BB
5		Explain various Debugging techniques	K2	2	Lecture	BB+ICT
6		Explain guidelines for debugging and role of debuggers	K2	1	Lecture	BB

Total 8

Unit-5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Describe the benefits of minimizing test suite and state reasons why the test suite grows.	K1	1	Lecture	BB
2		Explain types of test case prioritization.	K2	1	Lecture	BB
3		Illustrate the coverage based test case prioritization	K3	2	Lecture	BB+ICT
4		Explain risk-based and operational profile based test case prioritization	K2	2	Lecture	BB
5		Demonstrate test case prioritization using relevant slices.	K3	2	Lecture with Discussion and in class Assignment	BB+ICT
6		Assessing the effectiveness of a prioritized test suite.	K3	2	Lecture	BB

Total 10

Unit-6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO6	Define the concept of software quality and its factors	K1	1	Lecture	BB
2		Differentiate Quality control and Quality Assurance	K2	1	Lecture with Discussion and in class Assignment	BB
3		Explain various quality metrics	K2	1	Lecture	BB
4		Illustrate ISO 9126 SQA model	K2	1	Lecture	BB+ICT
5		Explain CMM model	K2	1	Lecture	BB
6		Define automated Testing and identify the need for automation	K1	1	Lecture	BB
7		Classify Testing tools	K2	1	Lecture	BB

8		Outline Guide lines for selecting a testing tool	K1	1	lecture	BB
9		List some commercial testing tools	K1	1	lecture	BB+ICT

Total 9

Total No. of Classes: 55

Machine Learning (PE-II)

Academic Year: 2020-21

Programme: B.Tech.

Year/Sem: VI

Name of the Course: Machine Learning **Course Code:** V18CST25

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S.No.	CO No.	Course Outcome	BTL
1.	1	Demonstrate basics of Machine Learning	K2
2.	2	Explain Various Classification Techniques	K2
3.	3	Explain Tree Based Learning and Ensemble Learning	K2
4.	4	Demonstrate Neural Networks and Multi Layer Perceptrons	K2
5.	5	Explain Dimensionality Reduction techniques	K2
6.	6	Demonstrate clustering algorithms	K2

Text Books:

1. Machine Learning: An Algorithmic Approach. Stephen Marsland, 2nd Edition, CRC press.
2. A First Course in Machine Learning; Volume in Machine Learning and Pattern Recognition Series – CRC-Taylor & Francis-Chapman & Hall Rogers S., Girolami M., (2011)

Reference Books:

1. Machine Learning: The art and Science of Algorithms that Make sense of Data. Peter Flach, Cambridge.
2. Machine Learning: Tom Mitchel, McGraw Hill Learning, 1997

Targeted Proficiency Level (For each course Outcome):

COs		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	55	55	55	55	55	55
	Level 1	50	50	50	50	50	50

Plan:

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos, & PSOs of the Programme		1	Discussion	ICT
2		Introduction to Machine Learning and Types of Machine Learning	K1	1	Lecture	ICT
3		Discuss Supervised Learning :Regression and Classification	K1	1	Lecture with Discussion	ICT
4		Illustrate The Machine Learning Process	K2	1	Lecture	ICT
5		Discuss the Terminology of ML: Weight Space, The Curse Of Dimensionality	K1	1	Lecture with Discussion	ICT
6		Demonstrate Knowing What You Know: Testing Machine Learning Algorithms	K2	2	Lecture	ICT
7		Demonstrate Some Basic Statistics. Averages Variance And Covariance. The Bias-Variance Trade-off	K2	2	Lecture	Chalk and Talk

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Discuss The General Problem, Probabilistic Classifiers: Bayes Classifier	K2	2	Lecture	ICT
2		Illustrate Probabilistic Classifiers: Logistic Regression	K2	2	Lecture	ICT
3		Illustrate Non-Probabilistic Classifiers: K-Nearest Neighbor	K2	3	Lecture	ICT
4		Illustrate Non-Probabilistic Classifiers:Support Vector Machines	K2	3	Lecture	ICT
5		Assessing classification performance	K2	2	Lecture with Discussion	ICT

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Describe Decision Trees: Construction of Decision Trees	K2	3	Lecture with Discussion	ICT
2		Illustrate the Classification And Regression Trees (CART)	K2	1	Lecture	ICT
3		Demonstrate Classification Example	K2	2	Lecture with Discussion	Chalk and Talk
4		Illustrate the Classification And Regression Trees (CART)	K2	1	Lecture with Discussion	ICT
5		Discuss about Bagging and Subbagging	K2	1	Lecture	ICT
6		Discuss about Random Forests	K2	1	Lecture	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Discuss about The Brain And The Neuron : Hebb's Rule	K1	1	Lecture	Chalk and Talk
2		Illustrate The Brain And The Neuron : Mcculloch And Pitts Neurons Limitations Of The Mcculloch And Pitts Model	K2	2	Lecture with Discussion	Chalk and Talk
3		Illustrate Neural Networks, Perceptron : The Learning Rate H, The Bias Input The Perceptron Learning Algorithm, An Example Of Perceptron Learning: Logic Functions Implementation	K2	3	Lecture	ICT, Chalk and Talk
4		Discuss about Linear Separability	K2	2	Lecture	ICT, Chalk and Talk
5		Discuss about Linear Regression	K2	2	Lecture	ICT, Chalk and Talk

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Discuss about Introduction to Dimensionality Reduction and Multi layer perceptron	K1	2	Lecture	ICT
2		Demonstrate Linear Discriminant Analysis (LDA)	K2	2	Lecture	ICT
3		Demonstrate Principal Components Analysis (PCA) , Kernel PCA	K2	2	Lecture with Discussion	ICT
4		Discuss about Factor Analysis	K2	1	Lecture with Discussion	ICT
5		Discuss about Independent Components Analysis (ICA)	K2	1	Lecture with Discussion	ICT
6		Discuss about Locally Linear Embedding	K2	2	Lecture	ICT

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Discuss about Unsupervised Learning:The K-Means Algorithm	K2	2	Lecture	ICT
2		Illustrate K-Means Algorithm :Dealing With Noise, K-Means Neural Network	K2	2	Lecture	ICT
3		Illustrate Normalisation, Better Weight Update Rule	K2	1	Lecture	ICT
4		Demonstrate The Iris Data set ,Using Competitive Learning For Clustering	K2	2	Lecture with Discussion	ICT
5		Discuss about Vector Quantisation	K2	1	Lecture	ICT

6		Discuss about The Self-Organising Feature Maps	K2	2	Lecture	ICT
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Total No. of Classes: 60

OOAD Through UML Lab Programme

LESSON PLAN

After completion of this course, the students will be able to:

Academic Year: 2020-21

Year/ Semester: VI

Name of the Course: OOAD THROUGH UML Lab Course Code: V18CSL08

Programme: B.Tech

Section: A,B,C& D

CO	Course Outcome	Knowledge Level
V18CSL08.1	Develop OOAD and UML concepts to identify Classes, Use Cases and their relationships	K3
V18CSL08.2	Develop Class diagrams	K3
V18CSL08.3	Develop Use case diagrams	K3
V18CSL08.4	Construct Interaction diagrams	K3
V18CSL08.5	Develop State chart, Activity diagrams	K3
V18CSL08.6	Develop component and deployment diagrams	K3

TEXT BOOKS:

1. The Unified Modeling Language User Guide, Grady Booch, James Rumbaugh, Ivar Jacobson, Pearson Education.

REFERENCE BOOKS:

1. UML 2 Toolkit, Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, WILEY-Dreamtech India Pvt. Ltd.
2. Fundamentals of Object Oriented Design in UML, Meilir Page-Jones, Pearson Education.
3. Modeling Software Systems Using UML2, Pascal Roques, WILEY- Dreamtech India Pvt. Ltd.

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Lecture Plan:

S. N O	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Develop OOAD and UML concepts to identify Classes, Use Cases and their relationships	K3	3	Lecture With Discussion	ICT
2	CO2	Construct the Class Diagram for Library management System, Railway reservation System.	K3	3	Lecture With Discussion	ICT
		Construct the Class Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
3	CO3	Construct the Usecase Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Usecase Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
4	C04	Construct the Interaction Diagrams for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Interaction Diagrams for Hospital Management system	K3	3	Lecture With Discussion	ICT
5	CO5	Construct the Activity diagrams for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Activity diagramfor Hospital Management system	K3	3	Lecture With Discussion	ICT
6	CO5	Construct the State chart for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the State chart Diagram for Hospital	K3	3	Lecture With	ICT

		Management system			Discussion	
7	CO6	Construct the Component Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Component Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT
8	CO6	Construct the Deployment Diagram for Library management System, Railway reservation System	K3	3	Lecture With Discussion	ICT
		Construct the Deployment Diagram for Hospital Management system	K3	3	Lecture With Discussion	ICT

Total Hours: 45

Technical Skills-IV

Academic Year: 2020-21

Programme:B.Tech.

Year/Sem: VI

Name of the Course: Technical Skills-IV

Course Code: V18CST63

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

CO	Course Outcomes	Knowledge Level
1	Demonstrate java fundamentals to solve real world computational problems.	K2
2	Illustrate object orientated concepts in solving problems with reusability feature.	K2
3	Apply collections on java to solve complex problems in linear time.	K3
4	Make use of divide and conquer approach and java collections to solve problems in linear and logarithmic time.	K3
5	Experiment with backtracking algorithm approach and java collections to solve complex problems in $O(n)$ or $O(n \log n)$.	K3
6	Develop programs to solve optimization, NP Hard problems using java collections and dynamic programming method.	K3

TEXT BOOKS:

1. Data structures and Algorithms made easy on Java: NarasimhaKarumanchi.
2. Thinking on Java - O'Reilly.
3. Java Complete Reference.

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5	CO6
Targeted Proficiency Level		60	60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60	60
	Level 2	50	50	50	50	50	50
	Level 1	40	40	40	40	40	40

Lecture Plan:

S.NO	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours required	Pedagogy	Teaching aids
1	CO 1	Dissemination of Department Vision, Mission, PEOs, POs, PSOs	-	-	-	
		DemonstrateControl Statements	K2	2	Lecture With Discussion	ICT
2		Illustrate Problem solving using Control Statements.	K2	6	Lecture With Discussion	ICT

Total Hours = 08

S.No	Course Outcome	Intended Learning Outcomes(ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	DemonstrateOOP Concepts.	K2	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using OOP Concepts	K2	10	Lecture with Discussion	ICT

Total Hours = 12

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	DemonstrateDivide& Conquer	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Divide and Conquer in linear time.	K3	6	Lecture with Discussion	ICT

Total Hours = 08

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	DemonstrateCollections.	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving usingCollection concept.	K4	10	Lecture with Discussion	ICT

Total Hours = 12

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Demonstrate Backtracking Concept and Greedy Method	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Backtracking	K3	8	Lecture with Discussion	ICT

Total Hours =10

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Demonstrate Optimization, Dynamic Programming	K3	2	Lecture with Discussion	ICT
2		Illustrate Problem solving using Collections and Dynamic Programming.	K3	8	Lecture with Discussion	ICT

Total Hours = 10

Total No. of Classes: 60

Internet of Things

Academic Year: 2020-21

Programme:B.Tech.

Year/Sem: VI

Name of the Course: Internet of Things

Course Code: V18ECTOE1

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

S. No.	CO No.	Course Outcome	Kx
1.	C319.1	Describe M2M and IOT Technologies.	K2
2.	C319.2	Identify the layers and protocols in IOT.	K2
3.	C319.3	Describe various communication technologies used in IOT.	K2
4.	C319.4	Demonstrate various hardware components required for IOT applications.	K2
5.	C319.5	Identify the cloud technologies.	K2
6.	C319.6	Explain the applications of IOT.	K2

Text Books:

1. Jan Holler, Vlasios Tsiatsis, Catherine Mulligan, Stefan Avesand, Stamatis Karnouskos, David Boyle, "From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence", 1st Edition, Academic Press, 2014.
2. IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things, David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Robert Barton, Jerome Henry, Cisco Press 800 East 96th Street Indianapolis, Indiana 46240 USA.
3. Vijay Madiseti and Arshdeep Bahga, "Internet of Things (A Hands-on- Approach)", 1st Edition, VPT, 2014.
4. Internet of Things: Architecture and Design Principles- By Raj Kamal, McGraw-Hill Education. 1st Edition, 2017.

Reference Books:

1. From Internet of Things to Smart Cities: Enabling Technologies - edited by Hongjian Sun, Chao Wang, Bashar I. Ahmad, CRC Press -2018.
2. Bernd Scholz-Reiter, Florian Michahelles, "Architecting the Internet of Things", ISBN 978-3- 642-19156-5 e-ISBN 978-3-642-19157-2, Springer.
3. IOT (Internet of Things) Programming: A Simple and Fast Way of Learning IOT, David Etter.

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	BTL	No. of Hours Required	Pedagogy	Teaching aids
		UNIT-1 INTRODUCTION [1]				
1.	CO1:	Describe the evolution of IoT from M2M	K1	1	Lecture with discussion	PPT and ICT
2.		Discuss Architectural Overview	K1	1	Lecture with discussion	PPT and ICT
3.		Illustrate the Building architecture	K2	1	Lecture with discussion	PPT and ICT
4.		Explain Main design principles and needed capabilities	K2	1	Lecture with discussion	PPT and ICT
5.		Outline the IoT architecture	K2	1	Lecture with discussion	PPT and ICT
6.		Explain M2M and IoT Technology Fundamentals: Devices	K2	2	Lecture with discussion	PPT and ICT
7.		Explain M2M and IoT Technology Fundamentals: Gateways.	K2	1	Lecture with discussion	PPT and ICT
		Total Classes		8		
		UNIT-2 IOT PROTOCOLS [2]	K2			
1.	CO2:	Describe the Functionality of Layers in IoT	K2	2	Lecture with discussion	PPT and ICT
2.		Illustrate IoT protocols: Wireless HART	K2	2	Lecture with discussion	PPT and ICT
3.		Illustrate IoT protocols: Z-Wave, 6LoWPAN	K2	3	Lecture with discussion	PPT and ICT
4.		Illustrate IoT protocols: RPL	K2	1	Lecture with discussion	PPT and ICT
5.		Illustrate IoT protocols: CoAP.	K2	2	Lecture with discussion	PPT and ICT
6.		Illustrate IoT protocols: MQTT	K2	2	Lecture with discussion	PPT and ICT
		Total Classes		12		
		UNIT-3 COMMUNICATION TECHNOLOGIES IN IOT [2, 4]	K2			
1.	CO3:	Explain IoT Connectivity standard: IEEE 802.15.4	K2	1	Lecture with discussion	PPT and ICT

2.		Explain IoT Connectivity technology: Wi-Fi	K2	2	Lecture with discussion	PPT and ICT
3.		Explain IoT Connectivity technology: Bluetooth	K2	2	Lecture with discussion	PPT and ICT
4.		Explain IoT Connectivity technology: Zigbee, LPWAN	K2	3	Lecture with discussion	PPT and ICT
5.		Explain IoT Connectivity technology: 5G Era.	K2	2	Lecture with discussion	PPT and ICT
		Total Classes		10		
	C04:	UNIT-4 SYSTEM HARDWARE [3, 4]	K2			
1.		Classify various Sensors and functionalities	K2	3	Lecture with discussion	PPT and ICT
2.		Classify various Actuators and functionalities	K2	2	Lecture with discussion	PPT and ICT
3.		Infer the Radio Frequency Identification concept	K2	1	Lecture with discussion	PPT and ICT
4.		Discuss the role of various Embedded Devices for IoT- RASPBERRY PI.	K2	5	Lecture with discussion	PPT and ICT
		Total Classes		11		
	C05:	UNIT-5 Cloud Computing [4]	K2			
1.		Describe the Data Collection concepts	K2	4	Lecture with discussion	PPT and ICT
2.		Illustrate the Storage and Computing Using a Cloud Platform for IoT Applications/Services.	K2	6	Lecture with discussion	PPT and ICT
		Total Classes		10		
	C06:	UNIT-6 IoT APPLICATIONS [2]	K2			
1.		Explain the real time applications of IoT - Smart and Connected Cities	K2	3	Lecture with discussion	PPT and ICT
2.		Explain the real time applications of IoT - Public Safety	K2	3	Lecture with discussion	PPT and ICT

3.		Explain the real time applications of IoT - Agriculture	K2	3	Lecture with discussion	PPT and ICT
		Total Classes		9		
Total Classes		60				

Data Mining Lab

Academic Year: 2020-21

Programme: B.Tech.

Year/Sem: VI

Name of the Course: Data Mining Lab

Course Code: V18CSL09

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

C319.1	Demonstrate Data Pre-processing techniques.[K3]
C319.2	Demonstrate Association Rule Mining techniques.[K3]
C319.3	Demonstrate Classification techniques. [K3]
C319.4	Demonstrate the Clustering techniques. [K3]

REFERENCE BOOKS:

1. Data Mining: Practical Machine Learning Tools and Techniques, Ian H. Witten, Eibe Frank, Mark A. Hall, 3rd Edition, Morgan Kaufmann Publishers
2. Data Mining Concepts and Techniques, Jiawei Han, Micheline Kamber, Jian Pei, 3rd Edition, Morgan Kaufmann Publishers
3. Introduction to Data Mining, Pang-Ning Tan, Michael Steinbach, Vipin Kumar, 1st Edition, Pearson Education Inc.

Lecture Plan:

S.N O	Course Outcom e	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours require d	Pedagogy	Teaching aids
1	CO 1	Demonstrate Data Preprocessing on predefined Weka dataset labor.arff	K3	3	Lecture With Discussion	ICT
		Create a student.arff dataset and Demonstrate Data Preprocessing on it	K3	3	Lecture With Discussion	ICT
2	CO2	Demonstrate Association rule process on predefined Weka dataset contactlenses.arff using apriori algorithm.	K3	3	Lecture With Discussion	ICT
		Create an employee.arff dataset and demonstrate Association	K3	3	Lecture With	ICT

		rule process on it using apriori algorithm			Discussion	
3	CO3	Demonstrate Classification process on student.arff dataset using j48 algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Classification process on employee.arff dataset using id3 algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Classification process on employee.arff dataset using Naïve Bayes algorithm	K3	3	Lecture With Discussion	ICT
		Create a customer.arff dataset and demonstrate Classification process on it using j48 algorithm	K3	3	Lecture With Discussion	ICT
4	C04	Demonstrate Clustering process on predefined Weka dataset iris.arff using simple k-means algorithm	K3	3	Lecture With Discussion	ICT
		Demonstrate Clustering process on dataset student.arff using simple k- means algorithm	K3	3	Lecture With Discussion	ICT

Total Hours: 30

Professional Communication Skills - III

Academic Year: 2020-21

Programme: B.Tech.

Year/Sem: VI

Name of the Course: Professional Communication Skills - III

Course Code: V18ENT03

Course Outcomes (Along with Knowledge Level):

After Completing the course Student will be able to:

After completion of this course, students will be able to:

CO1: Distinguish the subtle meanings of various words in different contexts, recognize similar words as well as words with contrast meanings and use them appropriately.[K2]

CO2: Interpret the passage using different strategies and answer the questions with ease.[K3]

CO3: Compare different pairs of words and draw analogy between the words. Choose an appropriate word to make a sentence meaningful.[K4]

CO4: Recognize the easiest and best possible way of solving problem in the area of Number and Letter Series, Analogy, Classification, Coding & Decoding Symbols, Ranking and Analytical Reasoning.[K1]

CO5: Investigate the different types of logics involved in Mirror and water Images, Logical Reasoning & Arithmetical Reasoning.[K4]

CO6: Find the common traps in the questions and errors likely to be made from the concepts of Blood Relations, Directions, Average, Clock and Calendar, Data Sufficiency, Permutations-Combinations and Probability.[K3]

Text Books:

T1: PIC-VOC

T2: VERBAL ABILITY-2

Reference Books:

- Sujani, T., et.al, 2016, Pic Voc, Sri Vasavi Engineering College.
- RAYMOND MURPHY, 2007, Essential Grammar in Use, Cambridge University press, 3rd University.
- Thomson, A.J. , 1997, A Practical English Grammar, Oxford Publications
- Alex, K. , 2010, Soft Skills, S.Chand Publications.
- Barrons, 2014, GRE-Barons, Galgotia Publications.
- Mohammed Muneer, 2004, CAT , Tata McGraw- Hill Education
- Helio Fred Garcia, 2012, The power of Communication, FT Press/Pearson Publications
- Work book-1 on Aptitude prepared by Training & Placement Cell, Sri Vasavi Engineering College.
- Magical Book on Quicker Maths-Tyra
- Practice Book on Quicker Maths-Kundan & Tyra

- R.S.Agarwal-Sultan Chand Publications
- R.S.Agarwal-Non Verbal Reasoning

Hyperlinks

1. <https://www.indiabix.com/>
2. <https://www.campusgate.co.in/>
3. <https://www.question paper.org/>

Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):

Course Outcome	Targeted Proficiency Level (% of Marks)	Targeted level of Attainment (% Students)
1	50	60
2	50	60
3	50	60
4	50	60
5	50	60
6	50	60

CO 1

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	To Make students understand vocabulary and its contextual usage.	K2	1	Lecture	PPT/A.V
2		Can guess the meaning of the word through the root.	K2	1	Discussion	A.V
3		Learn Synonyms-Antonyms and understand the subtle meanings of the group words.	K2	1	Lecture	PPT/A.V
4		Students learn and use new words	K2	1	Lecture	A.V
5		Can arrange sentences in sequence cohesively as per sequence	K2	1	Lecture	PPT/A.V

CO 2

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	Students can complete the sentence in a meaningful way.	K3	1	Lecture	PPT/A.V
2		The Format and significance of official mail is learned.	K3	1	Lecture	PPT/A.V
3		Can identify the topic sentence and construct a cohesive paragraph	K3	1	Lecture	PPT/A.V

CO 3

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	Can draw analogy between the words of different categories.	K4	1	Lecture	PPT/A.V
2		Comprehend the passage using different strategies.	K4	1	Lecture	PPT/A.V

CO 4

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	Students can Identify the Next letter or Number in a correct Relation.	K1	1	Lecture	PPT/A.V
2		Students can justify the relation between words and Numbers.	K1	1	Lecture	PPT/A.V
3		Students can identifying different one from group of terms.	K1	1	Lecture	PPT/A.V
4		Students can Describe their Rank in a class or in a Competition.	K1	1	Lecture	PPT/A.V

CO 5

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	Can Classify the ages in a family members/ Explain the relation between numbers.	K4	1	Lecture	PPT/A.V
2		Calculate the Actual time in Mirror and Water/ Classify the Images.	K4	1	Lecture	PPT/A.V
3		Differentiate the logic behind the conclusions.	K4	1	Lecture	PPT/A.V

4		Students can Explain the logic for given problem.	K4	1	Lecture	PPT/A.V
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CO 6

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 6	Students can choose the correct relation between the persons.	K3	1	Lecture	PPT/A.V
2		Students can show the correct direction.	K1	1	Lecture	PPT/A.V
3		Students can calculate the Average of data.	K1	1	Lecture	PPT/A.V
4		Students can Relate the correct day for given date and angle between two hands of a clock.	K1	2	Lecture	PPT/A.V
5		Students can Intercept data.	K1	1	Lecture	PPT/A.V
6		Students can report the Real Time Scenarios possibility..	K1	2	Lecture	PPT/A.V

Total No. of Classes:16